

Samyak Jain

Software & AI Developer

Canberra, ACT, Australia • 0415 412 545 • samyak1901@gmail.com

[linkedin.com/in/samyak-jain](https://www.linkedin.com/in/samyak-jain)

<https://github.com/samyak1901>

<https://samyak1901.github.io/website/>

PROFILE

Software and AI Developer with an Honours degree in Advanced Computing (Intelligent Systems), with experience across backend engineering, data platforms, and applied AI. Designs production services and data pipelines for biomedical research, public-health surveillance, and enterprise analytics, and builds AI-driven tools for search, analytics, and automated decision support, taking systems from early concept through to reliable, maintainable software.

EXPERIENCE

Software / AI Developer

Dec 2024 – Present

Pacific Analytics Pty Ltd

- Developed backend services, data platforms, and applied AI systems for biomedical research, bioinformatics automation, and public-health surveillance.
- Built a genomic cohort-builder platform that enables researchers to create, refine, save, analyse, and export study cohorts from large-scale sequencing metadata.
- Implemented dynamic faceted filtering, real-time cohort summaries, distribution analytics, and streaming data exports for research discovery workflows.
- Built Lumi, a multi-agent AI assistant for bioinformatics research, with sub-agent delegation, parallel execution, tool orchestration, permission controls, and full run-level auditability.
- Designed cross-session memory and context-management capabilities for Lumi, including durable fact extraction, relevance-based memory retrieval, and automatic context compaction.
- Designed a syndromic surveillance anomaly-detection engine using Z-score and EWMA models, adaptive thresholds, confidence scoring, demographic stratification, and geographic roll-up.
- Delivered secure multi-tenant REST APIs, asynchronous processing pipelines, and service-to-service integrations across the platform.

Data Engineer / Data Analyst

Jan 2021 – Nov 2024

smrtr

- Progressed from Junior Data Analyst to Data Engineer, delivering data engineering, analytics, and automation solutions for enterprise clients and internal data products.
- Built and maintained data pipelines that transformed raw, multi-source data into reliable analytical datasets, and developed SQL data models, stored procedures, and reporting layers for customer analytics and segmentation.
- Analysed large customer and behavioural datasets and built dashboards and visualisations for technical and non-technical stakeholders, and supported ML and analytics workflows through data preparation, feature engineering, validation, and performance analysis.

Undergraduate Research Assistant

Jul 2019 – May 2020

The Australian National University

- Developed machine-learning models for a predictive-modelling approach to identify magnesium alloys with optimal mechanical properties, supporting research into lightweight vehicle materials.

EDUCATION

Bachelor of Advanced Computing (Honours), Computer Science

The Australian National University • 2018 – 2021 • Specialisation: Intelligent Systems

Bachelor of Science, Computer Science

Manipal Academy of Higher Education • 2016 – 2018

CERTIFICATIONS

AWS — Machine Learning • Analyzing and Visualizing Data with Microsoft Power BI

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, JavaScript, Rust
AI & ML	LLM agents & tool orchestration (Vercel AI SDK), multi-agent systems, semantic memory / retrieval, anomaly detection
Backend & APIs	FastAPI, Elysia, REST microservices, Pydantic, SQLAlchemy, Drizzle ORM
Data	PostgreSQL, ClickHouse, Redis, pandas, NumPy
Platform & Security	Docker, Kubernetes, AWS, Celery, RabbitMQ, Keycloak (JWT, OAuth2 / M2M), OPA
BI & Visualisation	Microsoft Power BI, Tableau